# INORGANIC CUME

01-26-2018

(Note: Answer 5 of the 10 questions below. Only first 5 will be graded)

1. (***20 points***) Answer these questions, and show your chemistry to support each answer.
   - i) Why is the bond angle in $NH_3$ greater (or smaller) than (or same as) that in $PH_3$?
   - ii) Which one, among $CH_3^-$ and $NH_2^-$, has a stronger proton affinity than the other?
   - iii) Ranking the hydrated cations by their size: $Li^+$ (aq), $K^+$ (aq), $Cs^+$ (aq) and $Na^+$ (aq).
   - iv) Rank the LiF and CsI in the order of increasing the solubility in water.

2. (***20 points***) Based on the Latimer Diagram below,

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| *$E^o$(V)* | | *0.564* | | *1.44* | | *-0.70* | | *1.51* | | *-1.18* |
| | $MnO_4^-$ | → | $MnO_4^{2-}$ | → | $MnO_2$ | → | $Mn^{3+}$ | → | $Mn^{2+}$ | → | Mn |

   - i) Explain which Mn species can help convert $Cl^-$ to $Cl_2$, {*$E^o$=1.36 (V) for ½ $Cl_2 + e^- = Cl^-$*}
   - ii) Which steps might be reversible.

3. (***20 points***) Use a simple sketch to indicate –
   - i) a (222) lattice plane in an *fcc* crystal unit cell;
   - ii) how many atoms in a typical *fcc* crystal for a metal;

4. (***20 points***) Plotting the electrical resistivity against the temperature for a metal, for an insulator, for a semiconductor, and for a superconductor, respectively.

5. (***20 points***) Explain why a single-sheet of $(BN)_n$ can (or cannot) be as electrically conductive as a graphene.

6. (***20 points***) Use $Fe^{2+}$ and ligands of your choice to make a diamagnetic $O_h$ complex, and figure out the LFSE and # of unpaired electron(s) for the complex.

7. (***20 points***) Figure out the lowest energy term in the ground state atomic spectrum for the Ni in $Ni(CO)_4$ molecule.

8. (***20 points***) For the unstable $Cp_2Co$,
   - i) show your chemistry to explain why this compound is unstable;
   - ii) propose a reaction of your choice to turn this c compound to a more stable one.

9. (***20 points***) Find a molecule isoelectronic to borazine. Assign the point group for this molecule.

10. (***20 points***) Propose a stable organometallic complex that can form on the graphene surface, and a stable one on the $(BN)_n$ surface. Show your chemistry to support your proposal.